

STATEMENTS ARE CLASSIFIED INTO 5 DIFFERENT TYPES

- DATA DEFINITION LANGUAGE (DDL)
- DATA MANIPULATION LANGUAGE (DML)
- TRANSACTION CONTROL LANGUAGE (TCL)
- DATA CONTROL LANGUAGE (DCL)
- DATA QUERY LANGUAGE (DQL) -→ select, from, order by, group by, distinct

DATA DEFINITION LANGUAGE (DDL):

" DDL is used to construct an object in the database and deals with the Structure of the Object "

It has 5 statements:

1. CREATE
2. RENAME
3. ALTER
4. TRUNCATE
5. DROP

1. CREATE : " IT IS USED TO BUILD / CONSTRUCT AN OBJECT "

HOW TO CREATE/BUILD A TABLE :

1. Name of the table (OBJECT)
Tables cannot have same names .
2. Number of Columns .
3. Assign datatypes for the Columns. {MANDATORY }
4. Assign Constraints [NOT MANDATORY] .

EXAMPLE 1:

TABLE NAME: PRODUCT (OBJECT)

N0_OF COLUMNS: 4

COL_NAME	PID	PNAME	PRICE	DISCOUNT
DATA TYPE	NUMBER(2)	VARCHAR (10)	NUMBER(7,2)	NUMBER(6,2)
CONTARINTS	NOTNULL	NOTNULL	NOT NULL	NOT NULL
	UNIQUE			
			CHECK(PRICE>0)	
	PK			

```
CREATE TABLE PRODUCT (
PID NUMBER(2) PRIMARY KEY,
PNAME VARCHAR(10) NOT NULL,
PRICE NUMBER(7,2) NOT NULL CHECK(PRICE>0),
DISCOUNT NUMBER(7,2)
);
```

Syntax to create a table :

```
CREATE TABLE Table_Name (
Column_Name1 datatype constraint_type ,
Column_Name2 datatype constraint_type ,
Column_Name3 datatype constraint_type ,
.
.
Column_NameN datatype constraint_type
);
```

EXAMPLE 2:

TABEL NAME= CUSTOMER (OBJECT)
NO OF COL =5

CID	CNAME	PH_NO	ADDRESS	PID
NUMBER (3)	VARCHAR(15)	NUMBER(10)	VARCAHR(10)	NUMBER(2)
	NOT NULL	NOT NULL	NOT NULL	NOT NULL
		UNIQUE		
		CHECK(LENGTH (PH_NO=10))		
PK				
				FK

```
CREATE TABEL CUST (
CID NUMBER(3) PRIMARY KEY,
CNAME VARCHAR(15) NOT NULL,
PH_NO NUMBER(10) NOT NULL UNIQUE CHECK(LENGTH(PH_NO)=10),
ADDRESS VARCHAR(10) NOT NULL,
PID NUMBER(2) NOT NULL,
FOREIGN KEY (PID) REFERENCES PRODUCT(PID)
);
```

(FOREIGN KEY (COL_NAME) REFERENCES TABEL_NAME1 (COL_NAME))

OR

CONSTRAINT PID_FK FOREGIN KEY (PID) PRODUCT (PID)

2. **RENAME** : "IT IS USED TO CHANGE THE NAME OF THE OBJECT "

Syntax: RENAME OldTable_Name TO New_Name ;

RENAME CUST TO CUSTOMER;

3. **ALTER** : " IT IS USED TO MODIFY THE STRUCTURE OF THE TABLE "

○ **TO ADD A COLUMN :**

Syntax: ALTER TABLE Table_Name
ADD Column_Name Datatype Constraint_type

EXAMPLE: ALTER TABLE CUSTOMER
ADD EMAIL VARCHAR(15) UNIQUE;

○ **TO DROP A COLUMN :**

Syntax: ALTER TABLE Table_Name
DROP COLUMN Column_Name ;

EXAMPLE
ALTER TABLE CUSTOMER
DROP COLUMN EMAIL ;

○ **TO RENAME A COLUMN :**

Syntax: ALTER TABLE Table_Name
RENAME COLUMN Column_Name TO new_Column_Name ;

EXAMPLE:
ALTER TABLE PRODUCT
RENAME COLUMN DISCOUNT TO DISC;

○ **TO MODIFY THE DATATYPE :**

Syntax: ALTER TABLE Table_Name
MODIFY COLUMN_NAME New_Datatype;

○ EXAMPLE:
ALTER TABEL CUSTOMER
ALTER TABEL CUSTOMER

MODIFY ADDRESS VARCHAR(15);

○ **TO MODIFY NOT NULL CONSTRAINTS :**

Syntax: ALTER TABLE Table_Name
MODIFY COLUMN_NAME Existing_datatype [NULL]/NOT NULL;

Example : ALTER TABLE PRODUCT
MODIFY DISC NUMBER(7,2) Not Null ;

TRUNCATE : " IT IS USED TO REMOVE ALL THE RECORDS/ROWS/TUPLE FROM THE TABLE PERMANENTLY "

Syntax: TRUNCATE TABLE Table_Name ;

DROP: " IT IS USED TO REMOVE THE TABLE FROM THE DATABASE "

Syntax: DROP TABLE Table_Name ;

TO RECOVER THE TABLE:

Syntax: FLASHBACK TABLE Table_Name
TO BEFORE DROP ;

EXAMPLE

ADDRESS: BIN\$123QASSDAD

**BIN FOLDER
CUSTOMER**

FLASHBACK

**FLASHBACK TABLE CUSTOMER
TO BEFORE DROP;**

TO DELETE THE TABLE FROM BIN FOLDER :

Syntax: PURGE TABLE Table_Name ;

NOTE : DDL STATEMENTS ARE AUTO-COMMIT STATEMENTS

DATA MANIPULATION LANGUAGE (DML)

It is used to Manipulate the Object by performing insertion , updating and deletion .

1. INSERT
2. UPDATE
3. DELETE

1. **INSERT** : It is used to insert / create records in the table .

Syntax: INSERT INTO Table Name VALUES(v1 , v2 , v3);

PRODUCT:

PID	PNAME	PRICE	DISCOUNT
NUMBER(2)	VARCHAR(10)	NUMBER(7,2)	NUMBER(7,2)

EXAMPLE;

INSERT INTO PRODUCT VALUES(11, 'REALME' , 5000.00, 1000.00);

2. **UPDATE** :It is used to modify an existing value .

Syntax: UPDATE Table_Name
SET Col_Name = Value , Col_Name = Value
[WHERE stmt] ;

EXAMPLE:

CUSTOMER:

CID	CNAME	PH_NO	ADDRESS	PID
NUMBER(3)	VARCHAR(15)	NUMBER(10)	VARCHAR(15)	NUMBER(2)
1	DINGA	9874561231	BANGLORE	12
2	DINGI	9874654123	BANGLORE	22

EXAMPLE:

WAQTD UPADTE OF PHONE OF DINGA TO 1234567890

UPADTE CUSTOMER

SET PH_NO=1234567890, ADDRESS = 'MANGLORE'

WHERE CNAME='DINGA';

3. **DELETE** : It is used to remove a particular record from the table .

Syntax: **DELETE**
FROM Table_Name
[WHERE stmt];

CUSTOMER:

CID	CNAME	PH_NO	ADDRESS	PID
NUMBER(3)	VARCHAR(15)	NUMBER(10)	VARCHAR(15)	NUMBER(2)
1	DINGA	9874561231	BANGLORE	12
2	DINGI	9874654123	BANGLORE	22

EXAAMPLE:

WAQTD REMOVE DINGA FROM THE LIST OF CUSTOMER
DELETE FROM
CUSTOMER
WHERE
CNAME='DINGA';

3. TRANSACTION CONTROL LANGUAGE (TCL)

"It is used to control the transactions done on the database ".

The DML Operations performed on the Database are known as Transactions such as Insertion , Updating and Deletion .

Updating and Deletion .

We have 3 Statements :

1. COMMIT
2. ROLLBACK
3. SAVEPOINT

1.COMMIT : "This statement is used to SAVE the transactions into the DB".

Syntax: COMMIT ;

EXAMPLE:

QUERY



CONSOLE/SQL+

NAME	SAL



DATABASE

NAME	SAL

INERT INTO T1(V1,V2,V3)

NAME	SAL
A	100

NAME	SAL
A	100

COMMIT;
CREA

TE T1

AUTO

COM

MIT(D

DL)

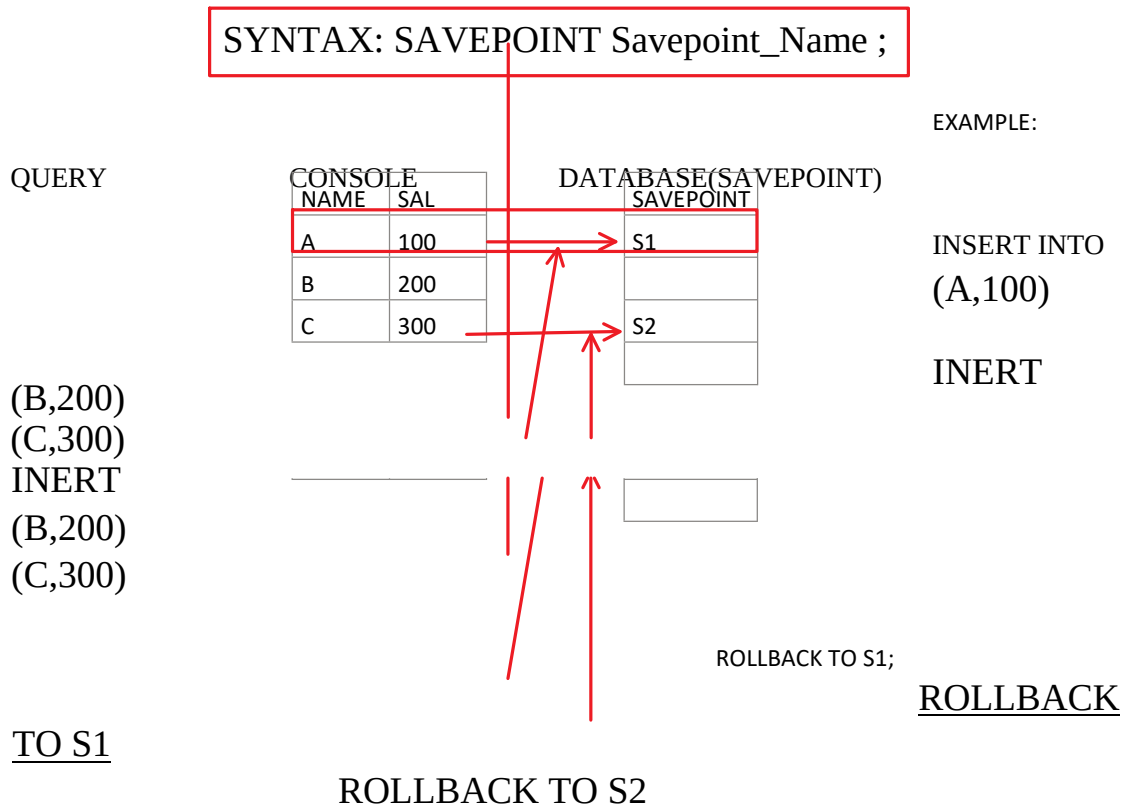
2.ROLLBACK :

This statement is used to Obtain only the saved data from the DB .
It will bring you to the point where you have committed for the last time .

SYNTAX: ROLLBACK ;

3. SAVEPOINT :

This statement is used to mark the positions or restoration points . (nothing related to DB) .



SYNTAX : ROLLBACK TO SAVEPOINT_NAME

4. DATA CONTROL LANGUAGE :

"This statement is used to control the flow of data between the users ".

We have 2 statements :

1. GRANT
2. REVOKE

1.GRANT : THIS STATEMENT IS USED TO GIVE PERMISSION TO A USER .

SYNTAX: GRANT SQL_STATEMENT

```
ON  
TABLE_NAME  
TO  
USER_NAME ;
```

2.REVOKE : THIS STATEMENT IS USED TO TAKE BACK THE PERMISSION FROM THE USER .

```
SYNTAX: REVOKE SQL_STATEMENT  
ON TABLE_NAME  
FROM USER_NAME ;
```

VIEW

"In SQL view is a virtual table based on the result set of an SQL statement"

SYNTAX

```
CREATE VIEW VIEW_NAME  
AS  
SELECT COLUMN_NAME  
FROM TABLE_NAME  
WHERE <CONDTION>
```

SYNTAX 2

```
DROP VIEW VIEW_NAME;
```

NOTE:

To give permission to create view !!

LOGIN AS SYSTEM

CONNECT ;

USER : SYSTEM

PWD: Oracle1234

GRANT CREATE VIEW TO SCOTT; (GIVING THE PERMISSION)

REVOKE CREATE VIEW FROM SCOTT; (TAKE BACK PERMISSON)

EXAMPLE :

EMPNO	ENAME	SAL	JOB
1	A	2000	MANAGER
2	B	1000	CLERK
3	C	5000	MANGER
4	D	1000	SLAESMAN